# Automated and Explainable Outcome Prediction in NSCLC: Beyond Manual Feature Extraction using LLM Embeddings

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## INTRODUCTION

➤ **Clinical notes** contain rich, unstructured **prognostic information**.

➤ **Manual extraction** is labor-intensive, slow, and **not scalable** for large cohorts.

➤ **LLM embeddings** can automate this process by generating **context-aware features**.

➤ **Explainability** is essential to translate AI predictions into **trustworthy clinical insights**.

## GOALS

➤ Automate **feature engineering** from **unstructured notes**.

➤ Predict **OS**, **PFS**, **DCR**, and **toxicity** across **IO, CHT, and IO +CHT**.

➤ Provide **sentence-level explainability** for clinical transparency.

➤ **Validate** AI logic via a rigorous **Fidelity framework**.

## DATASET

➤ **691 NSCLC patients**
- Istituto Nazionale dei Tumori

➤ **Apollo 11** project
- NCT05550961

➤ **Heterogeneous cohort** including patients treated with
- **Immunotherapy**
- **Chemotherapy**
- **Immuno-Chemotherapy**

➤ Multimodal analysis of
- unstructured **clinical reports**
- **laboratory results**
- **genomic profiling**

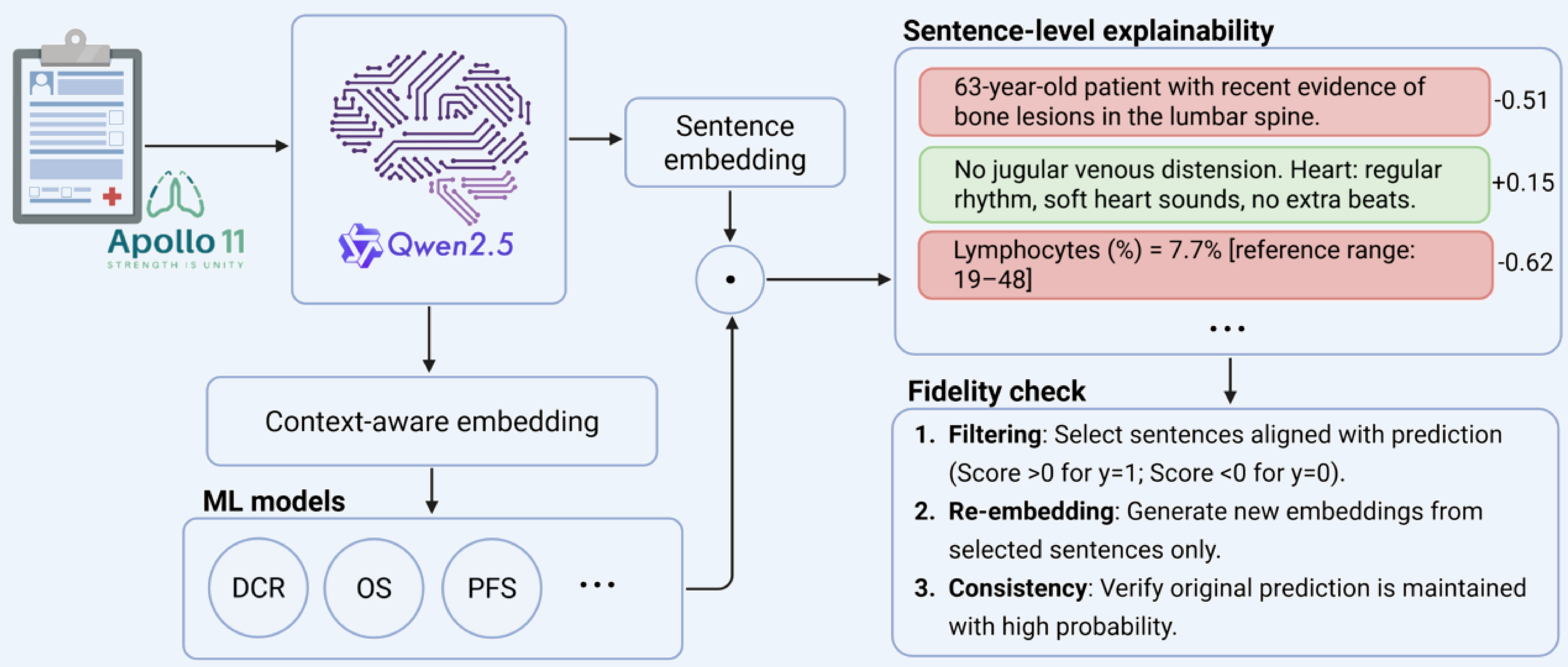
## METHODS

➤ **Qwen2.5-7B** generates patient-level embeddings

➤ Implemented on a proprietary **on-premise server** to ensure data privacy and **GDPR compliance**

➤ **Embeddings → prediction models**

➤ Sentence-level **explainability** + **fidelity**




## RESULTS

➤ Accurate prediction of **OS**, **PFS**, **DCR**, **toxicity**

➤ Explainability → key **clinical drivers**

➤ **Fidelity** validated explanations

➤ Fully **automated pipeline**

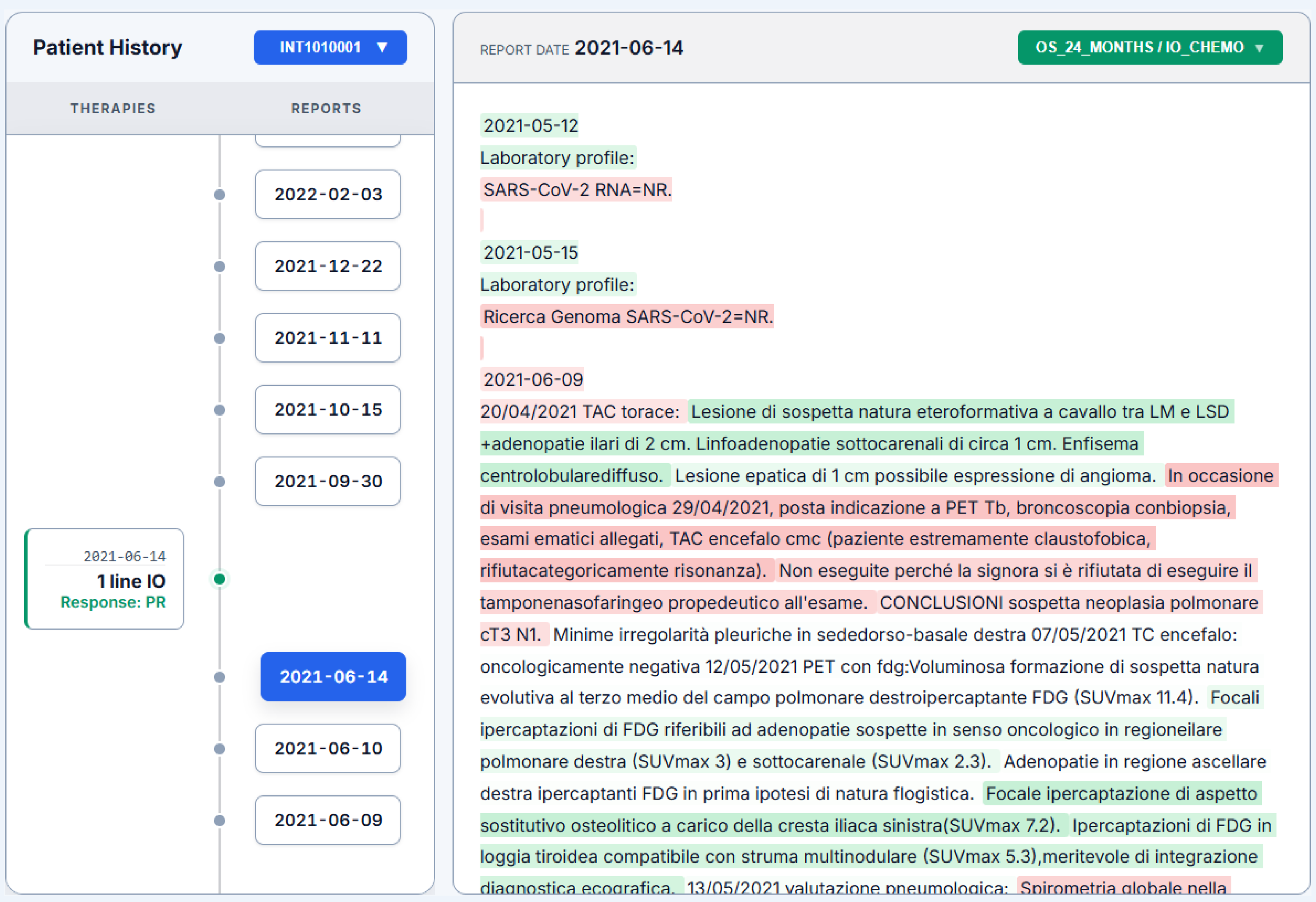
| Outcome | IO | CHT | IO+CHT |
| --- | --- | --- | --- |
| **OS 6** | $0.63 \pm 0.13$ | $0.79 \pm 0.19$ | $0.72 \pm 0.20$ |
| **OS 24** | $0.67 \pm 0.16$ | $0.81 \pm 0.19$ | $0.62 \pm 0.22$ |
| **DCR** | $0.55 \pm 0.14$ | $0.73 \pm 0.19$ | $0.56 \pm 0.21$ |
| **TOXICITY** | $0.57 \pm 0.15$ | * | $0.57 \pm 0.24$ |

**Table 1:** 5-fold CV AUC (Mean ± CI)

| Outcome | IO | CHT | IO+CHT |
| --- | --- | --- | --- |
| **OS** | $0.58 \pm 0.04$ | $0.68 \pm 0.06$ | $0.62 \pm 0.06$ |
| **PFS** | $0.61 \pm 0.08$ | * | $0.85 \pm 0.10$ |

**Table 2:** 5-fold CV C-index (Mean ± CI)

** Not assessed due to insufficient sample size for robust training.*




## CONCLUSION

➤ **LLM embeddings** replace manual feature engineering.

➤ **Transparent and reliable** clinical predictions.

➤ **Scalable** for real-world oncology.

**Next Steps:**

➤ **Longitudinal Analysis**: Capture disease evolution

➤ **Digital Twin**: Dynamic patient simulation

AI-ON-Lab

**Project details & contacts**